8.2 Integration by Parts

Technique for the integral of the form

$$\int f(x)\, g(x)\, dx,$$

where one part can be **differentiated** and another part can be **integrated** (easily).

Example: $\int x \sec^2 x \, dx$

to be differentiated. ↗ ↘ to be integrated

It is useful when f can be differentiated repeatedly to become zero and g can be integrated repeatedly without difficulty.

Example: $\int x^2 e^x \, dx$

choose x^2 to be differentiated

choose e^x to be integrated.

Product Rule in Integral Form

Recall that $\dfrac{d}{dx}f(x)g(x) = f(x)g'(x) + f'(x)g(x)$.

Integrating on both sides wrt x, we get

$$\int \frac{d}{dx}(f(x)g(x))\,dx = \int f(x)g'(x)\,dx + \int f'(x)g(x)\,dx$$

$$f(x)g(x) = \int f(x)g'(x)\,dx + \int f'(x)g(x)\,dx.$$

Integration by parts formula:

$$\int f(x)g'(x)\,dx = f(x)g(x) - \int f'(x)g(x)\,dx$$

If we let $u = f(x)$ and $dv = g'(x)\,dx$

we get $du = f'(x)\,dx$ $\quad v = g(x)$

$$\int u\,dv = uv - \int v\,du$$

hope is a simpler form

Alternative Form:

$$\int u\,dv = uv - \boxed{\int v\,du}$$

the hope is that this is a simpler form than the original integral.

The key to applying integration by parts is to break the integral to a suitable u and dv.

Example Using Integration by Parts

Find $\int x \cos x\,dx$.

Let $dv = \cos x\,dx$ and $u = x$.

$v = \sin x$ $\qquad du = dx$

$$\int x \cos x\,dx = uv - \int v\,du$$

$$= x \sin x - \int \sin x\,dx$$

$$= x \sin x + \cos x + C$$

There are four choices for u and dv in the above example. Namely, for $\int x \cos x \, dx$, we can choose

- $dv = x \cos x dx, u = 1$ → clearly no good!
- $dv = \cos x dx, u = x$ (✓)
- $dv = dx, u = x \cos x$

  $v = x$, $du = \cos x - x \sin x$

  $\Rightarrow \int x \cos x \, dx = x^2 \cos x - \underbrace{\int x(\cos x - x \sin x) \, dx}_{\text{more complicated}}$
- $dv = x dx, u = \cos x$

  $v = \frac{x^2}{2}$ $du = -\sin x$

  $\Rightarrow \int x \cos x \, dx = \frac{x^2}{2} \cos x + \underbrace{\int \frac{x^2}{2} \sin x \, dx}_{\text{more complicated}}$.

Example Integral of the Natural Logarithm

Find $\int \ln x \, dx$.

$u = \ln x \qquad dv = dx$

$\Rightarrow \quad du = \frac{1}{x} dx \quad , \quad v = x$

$$\Rightarrow \int \ln x \, dx = uv - \int v \, du$$

$$= x \ln x - \int x \cdot \frac{1}{x} \, dx$$

$$= x \ln x - \int 1 \, dx$$

$$= \underline{x \ln x - x + C}$$

Example

1. Find $\int \tan^{-1} x \, dx$.

2. Find $\int \sin^{-1} x \, dx$. (Exercise)

(1) $dv = dx \qquad u = \tan^{-1} x$

$v = x \qquad du = \frac{1}{1+x^2} dx$

$$\int \tan^{-1} x \, dx = x \tan^{-1} x - \int \frac{x}{1+x^2} dx$$

$w = 1+x^2 \Rightarrow dw = 2x\,dx$

$$= x \tan^{-1} x - \int \frac{dw/2}{w}$$

$$= x \tan^{-1} x - \frac{1}{2} \ln|w| + C$$

$$= x \tan^{-1} x - \frac{1}{2} \ln|1+x^2| + C$$

$$= x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) + C$$

Example Repeated Use of Integration by Parts

Evaluate $\int x^2 e^x \, dx$.

$u = x^2$, $dv = e^x \, dx$

$du = 2x \, dx$ $\quad v = e^x$

$$\int x^2 e^x \, dx = x^2 e^x - \int 2x e^x \, dx$$

$$= x^2 e^x - 2 \int x e^x \, dx$$

$u = x$ $\quad dv = e^x \, dx$

$du = dx$ $\quad v = e^x$

$$= x^2 e^x - 2 \left(x e^x - \int e^x \, dx \right)$$

$$= x^2 e^x - 2x e^x + 2 e^x + C$$

In general, we can use integration by parts to evaluate $\int x^n e^x \, dx$ for any positive integer n.

(We will see this again later.)

Example Solving for the Unknown Integral

Evaluate $\int e^x \cos x \, dx$.

Note: The alternate choice of $dv = e^x dx$ & $u = \cos x$ would work also.

$dv = \cos x \, dx \qquad u = e^x$

$\Rightarrow v = \sin x \qquad du = e^x dx$

$$\int e^x \cos x \, dx = e^x \sin x - \int e^x \sin x \, dx$$

Integration by parts again, but consistent choice as before.

$dv = \sin x \, dx, \quad u = e^x$

$v = -\cos x \qquad du = e^x dx$

$$= e^x \sin x - \left[-e^x \cos x + \int e^x \cos x \, dx \right]$$

$$= e^x (\sin x + \cos x) - \int e^x \cos x \, dx$$

$$\Rightarrow 2\int e^x \cos x \, dx = e^x (\sin x + \cos x) + C$$

$$\Rightarrow \int e^x \cos x \, dx = \frac{e^x}{2} (\sin x + \cos x) + C$$

Example Reduction Formula

Obtain a formula that expresses the integral

$$\int \cos^n x \, dx \quad , \; n \geq 2 .$$

in terms of an integral of a lower power of $\cos x$.

$$\int \cos^n x \, dx = \int (\cos^{n-1} x) \cos x \, dx$$

$u = \cos^{n-1} x \Rightarrow du = (n-1)\cos^{n-2} x (-\sin x)\, dx$.

$dv = \cos x \, dx \Rightarrow v = \sin x$.

$$= \sin x \cos^{n-1} x + \int (n-1)\cos^{n-2} x \sin^2 x \, dx .$$

$$= \sin x \cos^{n-1} x + (n-1)\int (1-\cos^2 x)\cos^{n-2} x \, dx \qquad (\because \sin^2 x = 1 - \cos^2 x)$$

$$= \sin x \cos^{n-1} x + (n-1)\int \cos^{n-2} x \, dx - (n-1)\int \cos^n x \, dx$$

lower power — original integral

$$n\int \cos^n x \, dx = \sin x \cos^{n-1} x + (n-1)\int \cos^{n-2} x \, dx .$$

$$\Rightarrow \boxed{\int \cos^n x \, dx = \frac{\sin x \cos^{n-1} x}{n} + \frac{n-1}{n}\int \cos^{n-2} x \, dx}$$

$$\int \cos^n x\,dx = \frac{\sin x \cos^{n-1} x}{n} + \frac{n-1}{n}\int \cos^{n-2} x\,dx$$

$$\int \cos^3 x\,dx = \frac{\sin x \cos^2 x}{3} + \frac{2}{3}\int \cos x\,dx \qquad n=3$$

$$= \frac{\sin x \cos^2 x}{3} + \frac{2}{3}\sin x + C$$

$$\int \cos^4 x\,dx = \frac{\sin x \cos^3 x}{4} + \frac{3}{4}\int \cos^2 x\,dx \qquad n=4$$

→ apply formula with $n=2$

$$= \frac{\sin x \cos^3 x}{4} + \frac{3}{4}\left[\frac{\sin x \cos x}{2} + \frac{1}{2}\int 1\,dx\right]$$

$$= \frac{\sin x \cos^3 x}{4} + \frac{3}{8}\sin x \cos x + \frac{3}{8}x + C$$

Example: Substitution + By Parts

- $\int e^{\sqrt{2x-1}}\,dx =$

- $\int \cos(\ln x)\,dx =$

① $\int e^{\sqrt{2x-1}}\,dx = \int w\,e^w\,dw =$

$w = \sqrt{2x-1}$

$dw = \frac{1}{x\sqrt{2x-1}} \cdot x\,dx$

$= \frac{1}{w}\,dx$

$dx = w\,dw$

$u = w \quad dv = e^w\,dw$

$du = dw \quad v = e^w$

$= we^w - \int e^w\,dw$

$= we^w - e^w + C$

$= \sqrt{2x-1}\,e^{\sqrt{2x-1}} - e^{\sqrt{2x-1}} + C$

② $\int \cos(\ln x)\,dx = \int e^w \cos(w)\,dw$

$w = \ln x$

$dw = \frac{1}{x}\,dx$

$dx = x\,dw = e^w\,dw$

$= \frac{e^w}{2}(\cos w + \sin w) + C$ (from previous example)

$= \frac{x}{2}(\cos(\ln x) + \sin(\ln x)) + C$

Evaluating Definite Integrals by Parts

Integration by Parts Formula for Definite Integrals

- $\int_a^b f(x)g'(x)dx = [f(x)g(x)]_a^b - \int_a^b f'(x)g(x)dx$
- $\int_a^b udv = [uv]_a^b - \int_a^b vdu$

Example

Evaluate $\int_0^4 xe^{-x}\,dx$: $[-xe^{-x}]_0^4 - \int_0^4 (-e^{-x})\,dx$

$u = x \Rightarrow du = dx$

$dv = e^{-x}dx$

$v = -e^{-x}$

$= -4e^{-4} + \int_0^4 e^{-x}\,dx$

$= -4e^{-4} + [-e^{-x}]_0^4$

$= -4e^{-4} + (-e^{-4} + 1)$

$= \underline{1 - 5e^{-4}} \approx 0.91$

Evaluate $\int_{2/\sqrt{3}}^{2} x\sec^{-1}x\,dx$. *Ans:* $\frac{5\pi}{9} - \frac{1}{2}\left(\sqrt{3} - \frac{1}{\sqrt{3}}\right)$

$dv = x\,dx \Rightarrow v = \frac{x^2}{2}$

$u = \sec^{-1}x$

$\Rightarrow du = \frac{1}{|x|\sqrt{x^2-1}}\,dx$

$$= \left[\frac{x^2}{2}\sec^{-1}x\right]_{2/\sqrt{3}}^{2} - \int_{2/\sqrt{3}}^{2} \frac{x^2}{2}\cdot\frac{1}{|x|\sqrt{x^2-1}}\,dx$$

$$= \left(2\sec^{-1}2 - \frac{2}{3}\sec^{-1}\left(\frac{2}{\sqrt{3}}\right)\right) - \int_{2/\sqrt{3}}^{2} \frac{x/2}{\sqrt{x^2-1}}\,dx$$

$w = x^2 - 1$

$dw = 2x\,dx$

$$= 2\frac{\pi}{3} - \frac{2}{3}\frac{\pi}{6} - \int_{1/3}^{3} \frac{dw/4}{\sqrt{w}}$$

when $x = 2 \Rightarrow w = 3$

$x = \frac{2}{\sqrt{3}} \Rightarrow w = \frac{1}{3}$

$$= \frac{5\pi}{9} - \left[\frac{1}{2}\sqrt{w}\right]_{1/3}^{3} = \frac{5\pi}{9} - \left(\frac{1}{2}\cdot\sqrt{3} - \frac{1}{2}\frac{1}{\sqrt{3}}\right)$$

$$= \frac{5\pi}{9} - \frac{1}{2}\left(\sqrt{3} - \frac{1}{\sqrt{3}}\right)$$

Tabular Integration

For integration by parts with many repetitions.

A good way to organise the calculations.

$$\int f(x)\, g(x)\, dx$$

We want to be able to differentiate $f(x)$ repeatedly to become zero, and we want to be able to integrate $g(x)$ repeatedly (without much difficulty).

EXAMPLE 7 Evaluate

$$\int x^2 e^x \, dx.$$

Solution With $f(x) = x^2$ and $g(x) = e^x$, we list:

$f(x)$ and its derivatives		$g(x)$ and its integrals
x^2	(+) →	e^x
$2x$	(−) →	e^x
2	(+) →	e^x
0	→	e^x

$$\int x^2 e^x \, dx = x e^x - 2x e^x + 2e^x + C$$

Example Evaluate $\int x^4 e^{-2x}\,dx$ using tabular integration.

Let $f(x) = x^4$ and $g(x) = e^{-2x}$.

$f(x)$ and its derivatives	$g(x)$ and its integrals
$f(x) = x^4$ (u)	$g(x) = e^{-2x}$ (dv)
$f'(x) = 4x^3$	$-\frac{1}{2}e^{-2x}$
$f''(x) = 12x^2$	$\frac{1}{4}e^{-2x}$
$f'''(x) = 24x$	$-\frac{1}{8}e^{-2x}$
$f^{(4)}(x) = 24$	$\frac{1}{16}e^{-2x}$
$f^{(5)}(x) = 0$	$-\frac{1}{32}e^{-2x}$

(+), (−), (+), (−), (+)

differentiate

integrate

$$\int x^4 e^{-2x}\,dx = -\frac{1}{2}x^4 e^{-2x} - x^3 e^{-2x} - \frac{3}{2}x^2 e^{-2x} - \frac{3}{2}x e^{-2x} - \frac{3}{4}e^{-2x} + C$$

$$= -e^{-2x}\left(\frac{1}{2}x^4 + x^3 + \frac{3}{2}x^2 + \frac{3}{2}x + \frac{3}{4}\right) + C$$

Example:

$$\int \sqrt{x}\sin^{-1}\sqrt{x}\,dx = uv - \int v\,du$$

$u = \sin^{-1}\sqrt{x}$ $\quad dv = \sqrt{x}\,dx$

$v = \frac{2}{3}x^{\frac{3}{2}}$

$du = \frac{1}{\sqrt{1-x}}\frac{d}{dx}(\sqrt{x})$

$= \frac{1}{2\sqrt{x}\sqrt{1-x}}\,dx$

$= \frac{2}{3}x^{\frac{3}{2}}\sin^{-1}\sqrt{x} - \int \frac{2}{3}x^{\frac{3}{2}}\frac{1}{2\sqrt{x}\sqrt{1-x}}\,dx$

$= \frac{2}{3}x^{\frac{3}{2}}\sin^{-1}\sqrt{x} - \frac{1}{3}\int \frac{x}{\sqrt{1-x}}\,dx$.

$w = \sqrt{1-x} \Rightarrow x = 1-w^2$

$dw = -\frac{1}{2\sqrt{1-x}}\,dx$

$= -\frac{1}{2w}\,dx$

$= \frac{2}{3}x^{\frac{3}{2}}\sin^{-1}\sqrt{x} - \frac{1}{3}\int \frac{1-w^2}{w}(-2w\,dw)$

$= \frac{2}{3}x^{\frac{3}{2}}\sin^{-1}\sqrt{x} + \frac{2}{3}\int (1-w^2)\,dw$

$= \frac{2}{3}x^{\frac{3}{2}}\sin^{-1}\sqrt{x} + \frac{2}{3}w - \frac{2}{9}w^3 + C$

$= \frac{2}{3}x^{\frac{3}{2}}\sin^{-1}\sqrt{x} + \frac{2}{3}\sqrt{1-x} - \frac{2}{9}(1-x)^{\frac{3}{2}} + C$

